

DDA: Dynamic Device Allocation — Latency Pool + Throughput Pool

Llama2-70B | 80 Decoder Layers | embed=8192 | 32 total devices split: N_L=12 + N_T=20

[L] Pool L — Latency Pool

N_L = 12 devices | TP = 12 | PP = 1

Interactive Request (5s relaxed SLO)



Broadcast → parallel GEMV (each holds 1/12 FC weight rows) → All-Reduce (CXL)

PP = 1: All 80 Decoder Layers processed in ONE stage

Layers 1 - 80 (all in one stage, processed by TP-12 group)

1 token in-flight (PP=1, no pipeline depth)



TTFT = 3.363 s ✓ Meets 5s SLO (1s SLO not achievable for 70B)

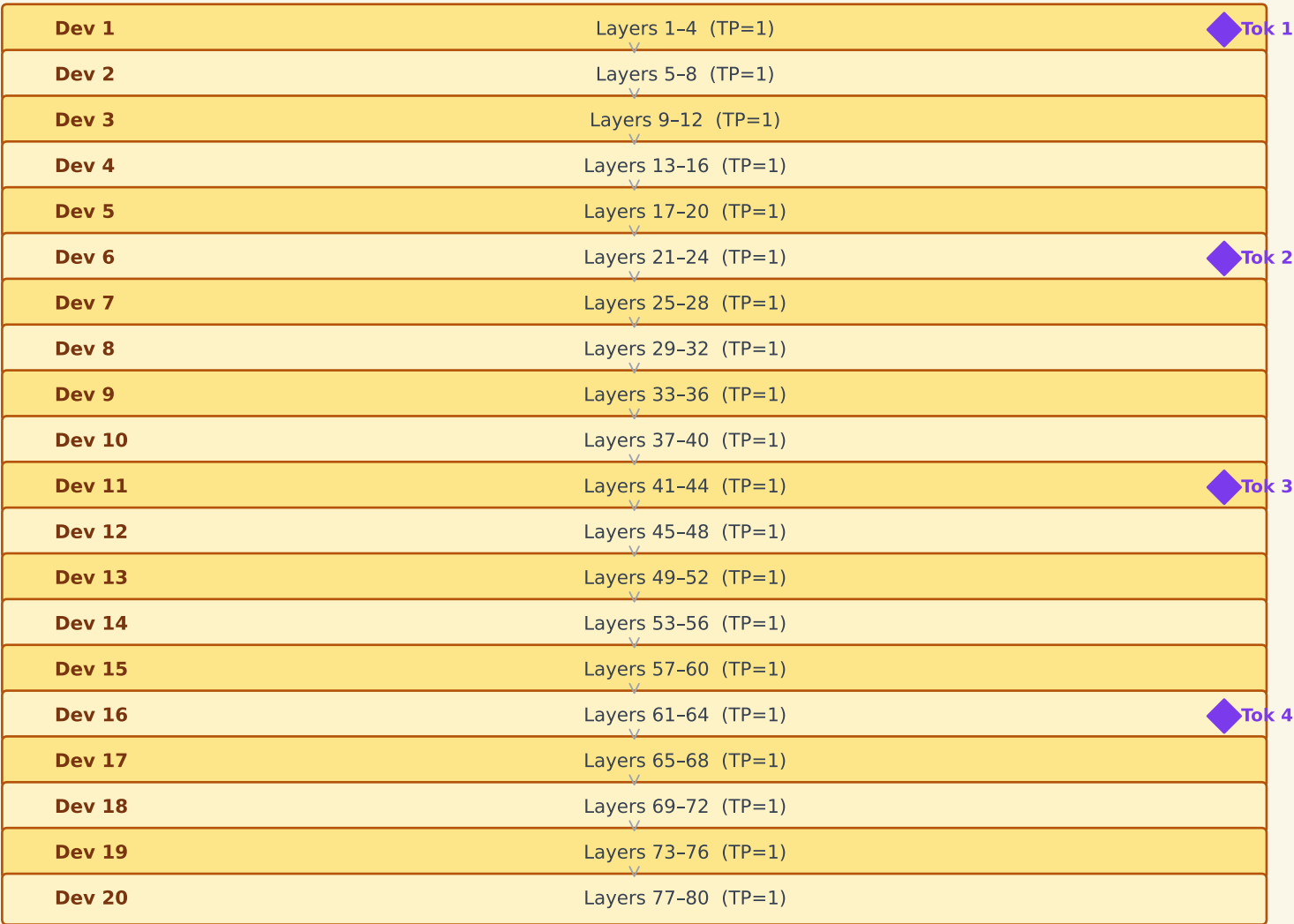
Pool L tput = 101.6 tok/s (low — single-token serving)

High TP → each token uses all 12 devices → best achievable latency

[T] Pool T — Throughput Pool

N_T = 20 devices | TP = 1 | PP = 80

Batch Requests (high-throughput workload)



4 tokens in-flight simultaneously (pipeline parallelism)

Pool T tput = 955.8 tok/s (high pipeline throughput)

TTFT >> 5s (batch / offline workloads only)

DDA System Throughput = Pool_L + Pool_T = 101.6 + 955.8 = 1057 tok/s vs Best static (PP=2, TP=16): 216 tok/s → 4.9× gain